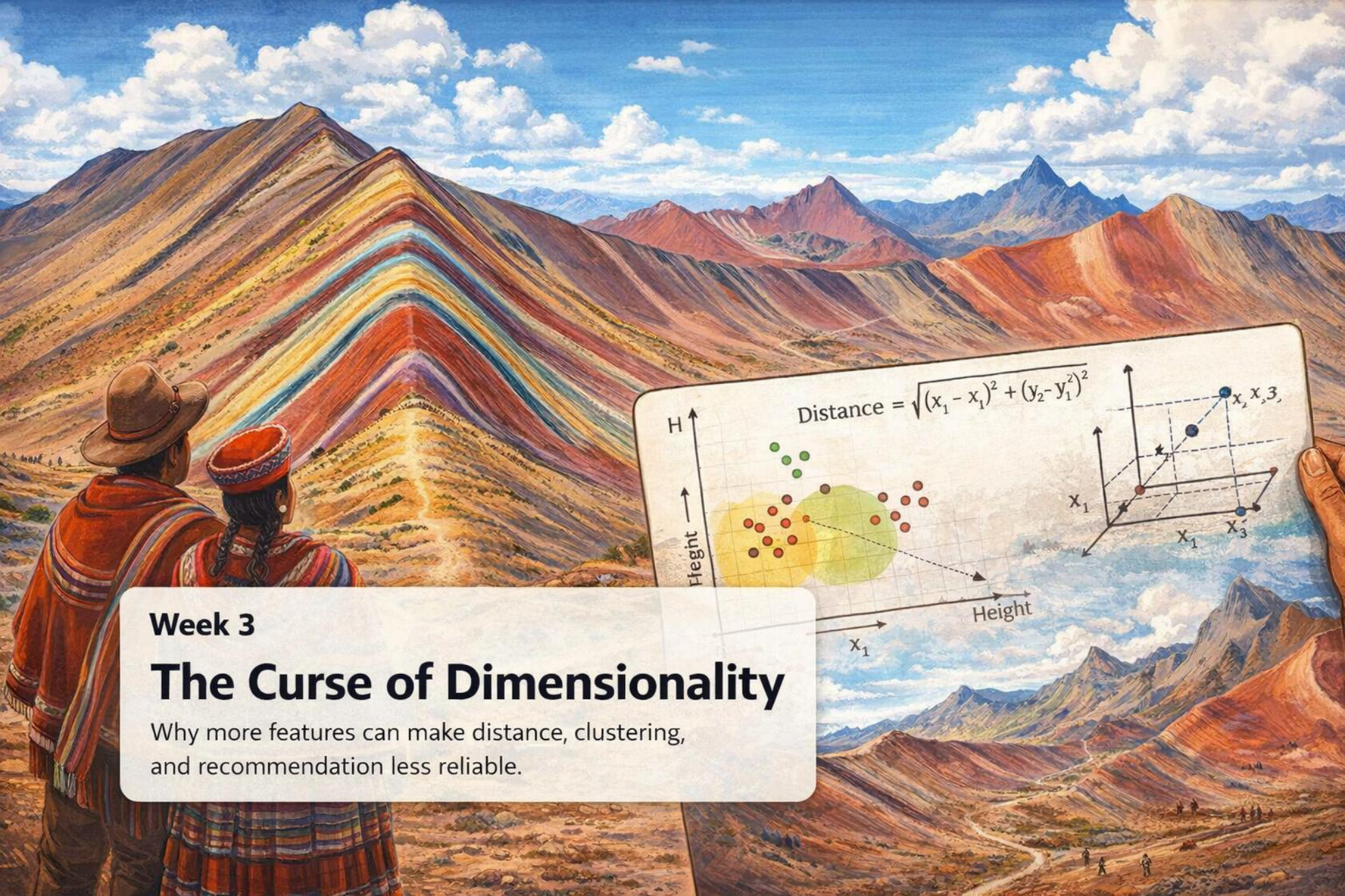
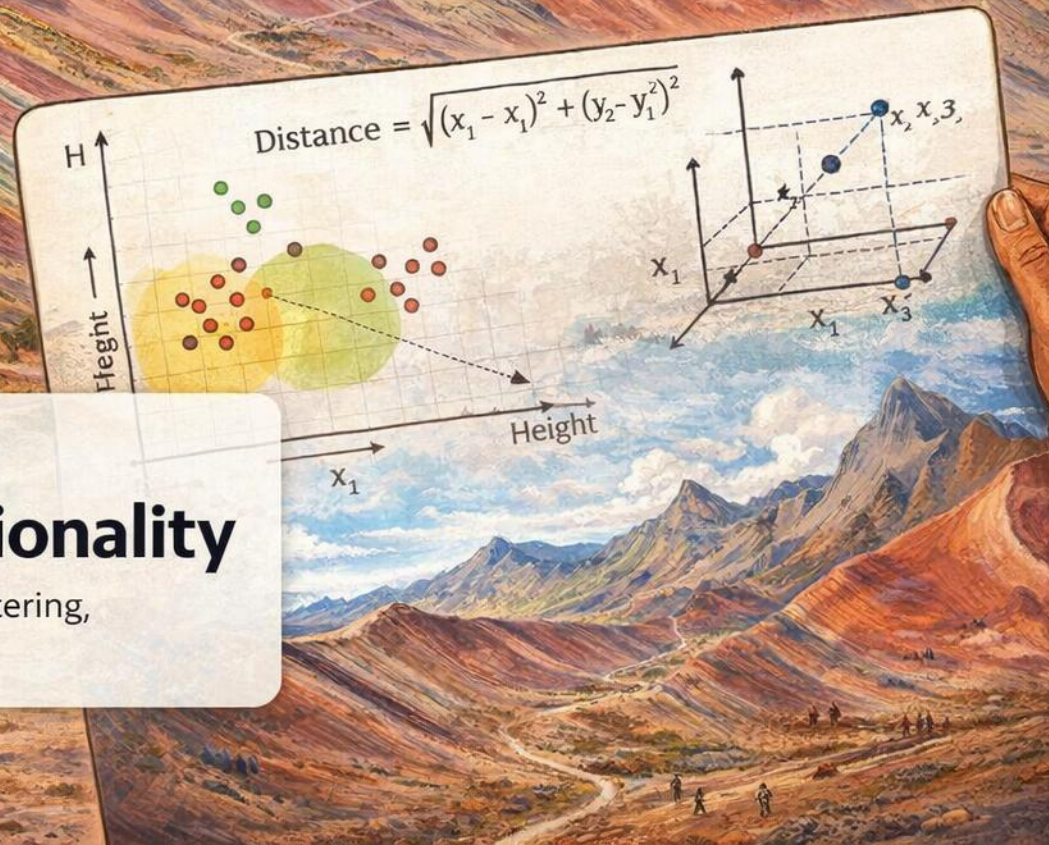




Week 3

The Curse of Dimensionality

Why more features can make distance, clustering, and recommendation less reliable.



Mathias expands the feature space again.

I made a feature vector with tempo, energy, danceability, acousticness, lyric counts, n-grams, embeddings, playlist statistics, and 2,000 extra columns just in case.



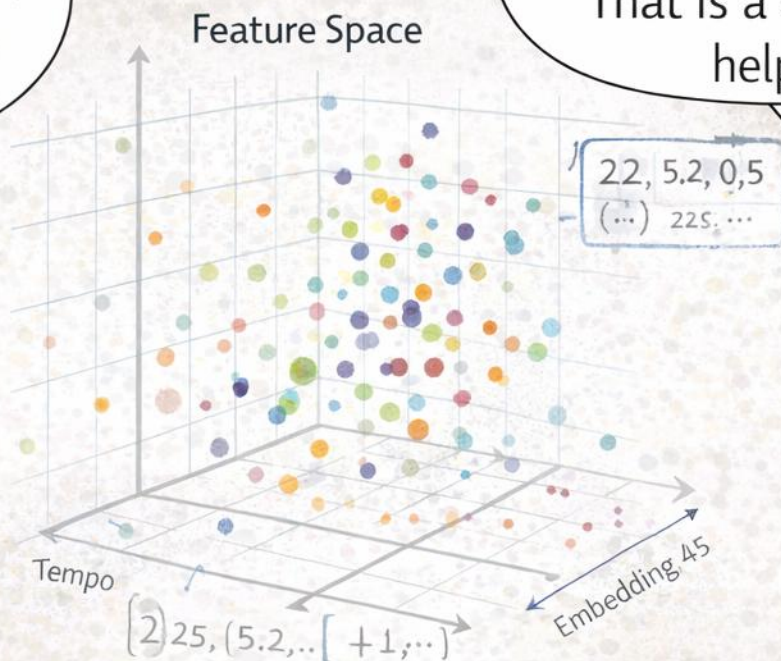
Feature-space dialogue

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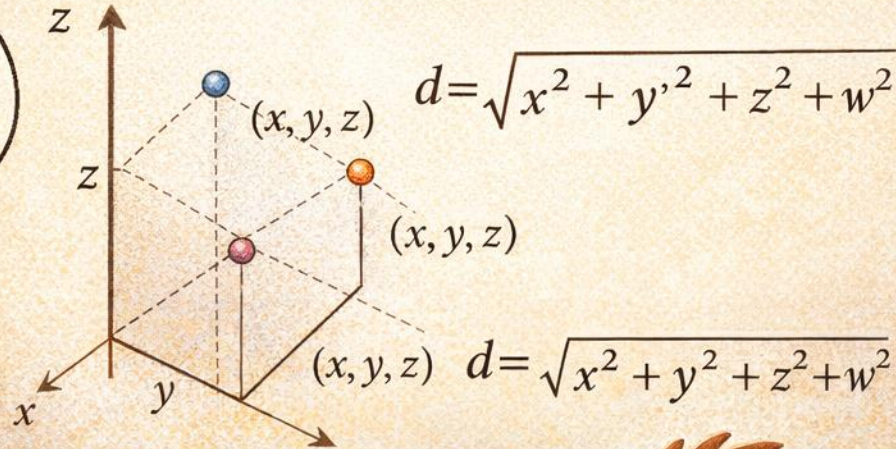
That is not a feature space. That is a cry for help.



More dimensions means more nuance.

Excellent. More dimensions means more nuance.

Excellent. More dimensions means more nuance.



Distance becomes less useful

And also more ways for distance to become useless.

And also more ways
for distance to
become useless.

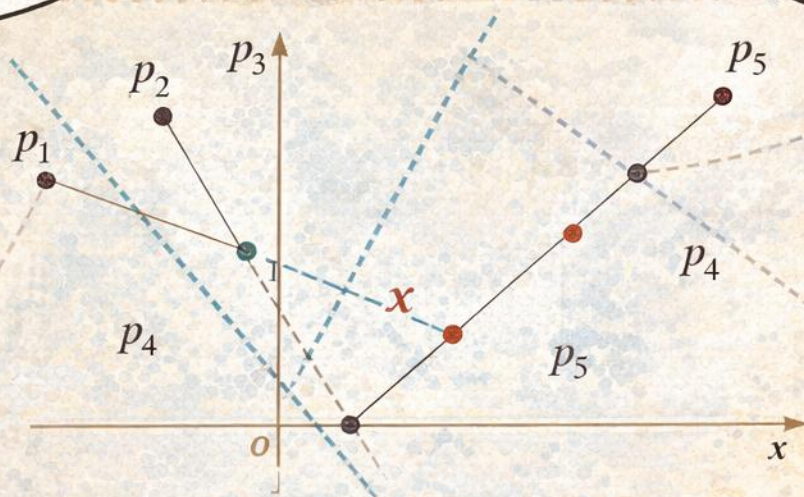


Geometry harms correctly.

. Context: Now you are finally being harmed by geometry in the correct way. | So the nearest neighbor may not be near?

Now you are finally being harmed by geometry in the correct way.

So the nearest neighbor may not be near?



$$(x - p_1)^T (x - p_1) < (x - p_5)^T (x - p_5))$$



More variables do not guarantee more usable information

PachaMix audio has 518 numeric descriptors; lyric spaces are even larger.

song	tempo	energy	dance	lyrics	embed_1	embed_2	...
Track A	128	0.83	0.71	142	0.31	-0.22	0.05
Track B	91	0.42	0.37	601	-0.07	0.44	-0.19
Track C	147	0.91	0.88	88	0.54	-0.38	0.11

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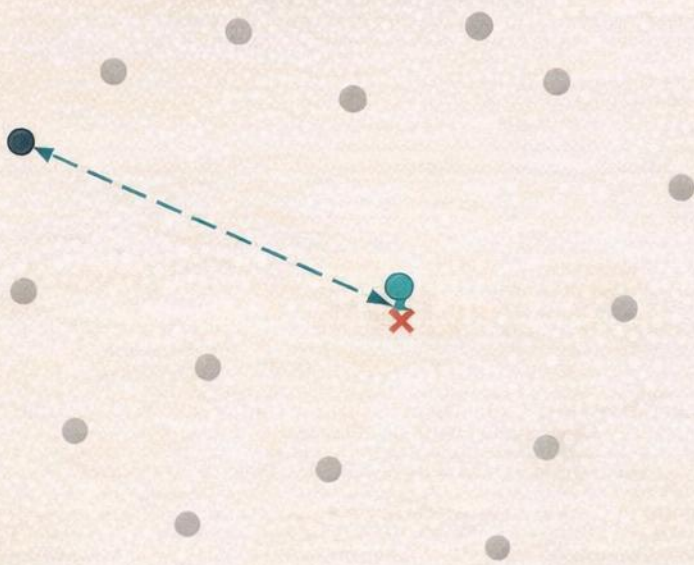
More columns can add:

- noise
- sparsity
- distance instability
- sample complexity

Distance intuition fails in high dimension

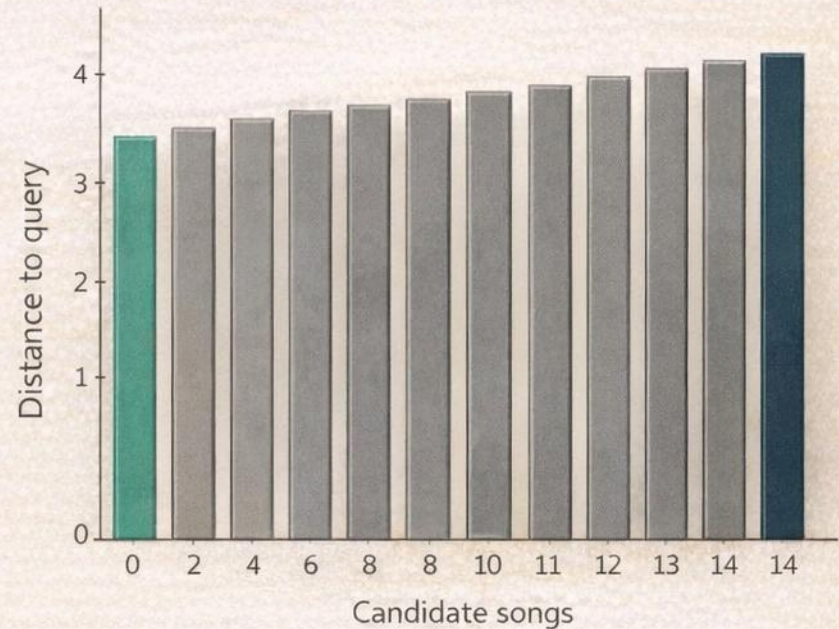
The nearest and farthest candidates stop looking meaningfully different.

2D: close and far are visually distinct



In 2D the query has a visibly close neighbor.

100D: distances bunch together



In 100D the sorted distances bunch together.

Hypercube versus hypersphere

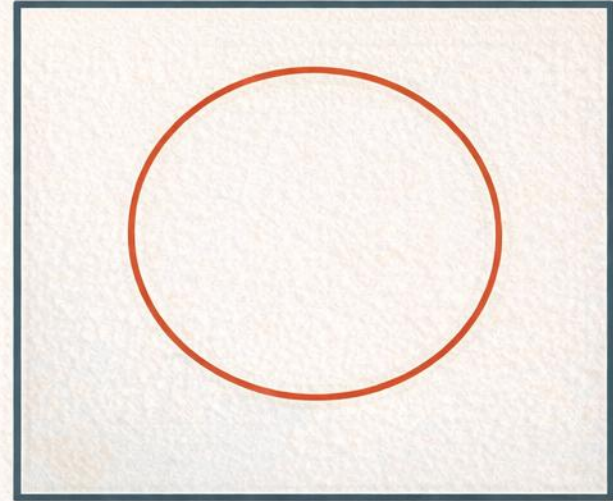
The geometry itself changes as dimension grows.

$$V_d(1) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)}$$

$$\text{Vol}([-1, 1]^d) = 2^d$$

$$\frac{V_d(1)}{2^d} = \frac{\pi^{d/2}}{2^d \Gamma(d/2 + 1)} \rightarrow 0$$

The d-dimensional unit ball occupies a vanishing fraction of the surrounding cube as dimension grows.

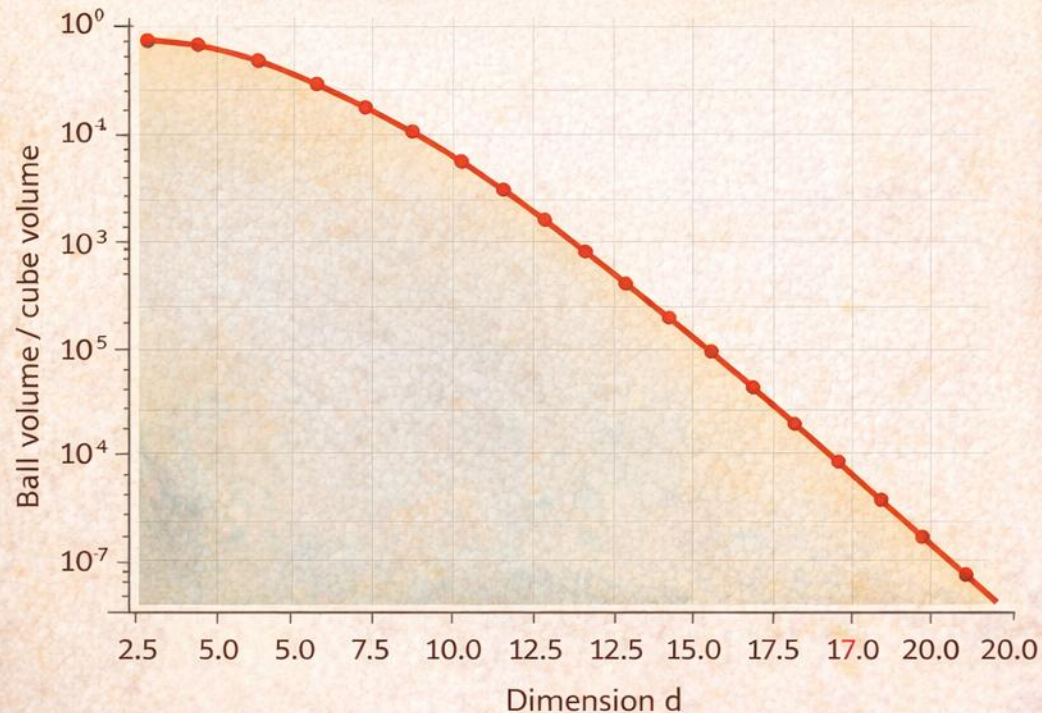


Even an inscribed ball occupies a vanishing share of the cube.

The inscribed ball vanishes inside the cube

Most of the ambient volume moves away from the center.

Inscribed ball share vanishes quickly



As dimension increases, local neighborhoods stop behaving like the ones we imagine from 2D or 3D drawings. That is why 'near' becomes harder to interpret.

Expected squared distance in the unit hypercube

The first quantitative sign of the curse is that distances expand with dimension.

$$x, y \sim \text{Unif}([0, 1]^d)$$

$$\begin{aligned} E\|x - y\|^2 &= \sum_{j=1}^d E(x_j - y_j)^2 \\ &= d \cdot \frac{1}{6} = \frac{d}{6} \end{aligned}$$

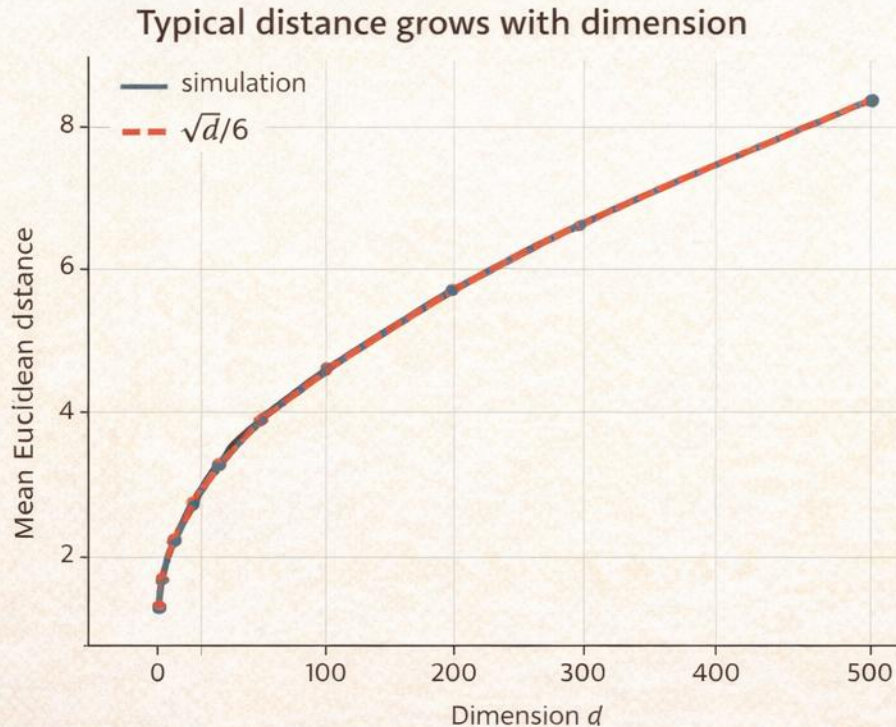
Each coordinate contributes 1/6 in expectation, so squared distances add linearly with d .

That is why the Euclidean distance itself typically scales like $\sqrt{d}$.

The derivation is simple but important: squared distances add coordinate by coordinate. More dimensions therefore stretch the ambient space even before we discuss nearest neighbors.

Expected distance grows with dimension

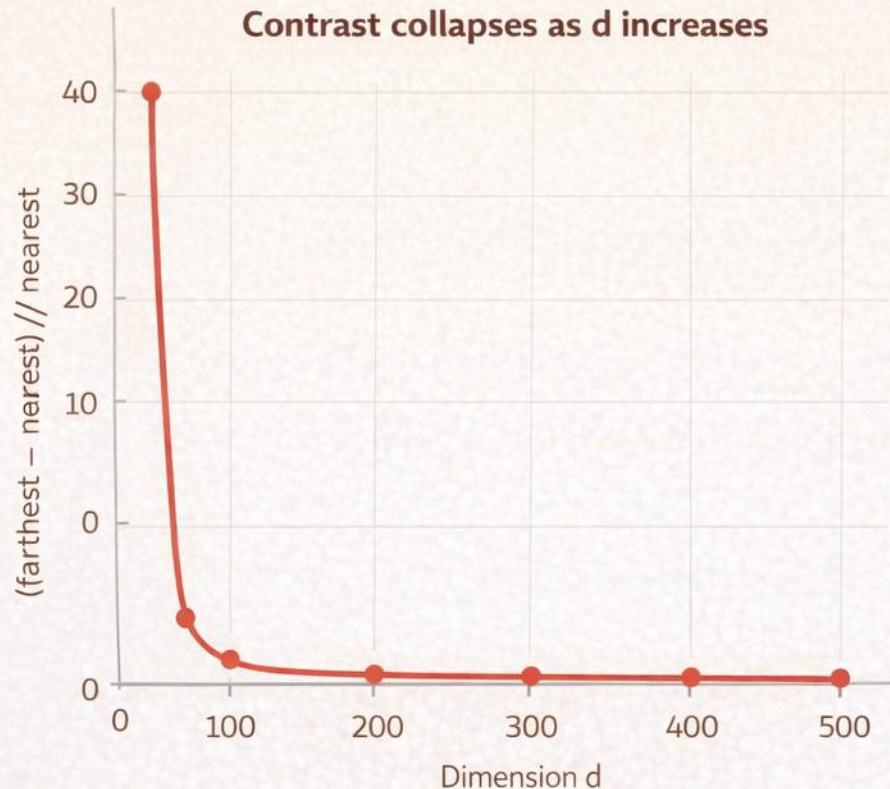
Simulation follows the theoretical $\sqrt{d}$ scaling very closely.



Distances get larger on average, but that alone is not the deepest problem. The harder part is that relative contrast shrinks while computation gets harder.

Distance concentration heuristic

Nearest and farthest candidates become less distinguishable in relative terms.



$$\frac{\max_i d(x, x_i) - \min_i d(x, x_i)}{\min_i d(x, x_i)} \rightarrow 0$$

This heuristic is the curse in operational form: the nearest and farthest songs become relatively indistinguishable.

When contrast collapses, distance ranking stops being a strong source of information.

Why recommendation, KNN, and clustering suffer

Distance-based methods depend on neighborhoods that still carry contrast.

K-nearest neighbors



Nearest is only marginally closer than the rest, so the ranking becomes fragile.

Clustering



Clusters depend on weak distance contrast, noise, and preprocessing choices.

Recommendation



A 'similar song' may be a metric artifact rather than a meaningful neighbor.

CODE

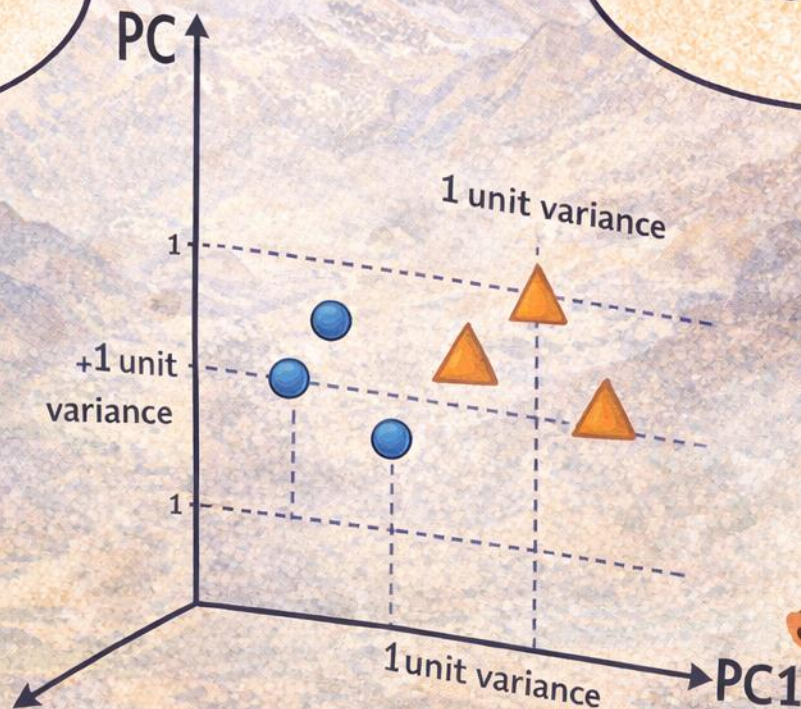
SLEEP

REPEAT

Standardized but still misleading

Good. Now the distances are consistently misleading. | But we standardized the features.

Good. Now the distances are consistently misleading.

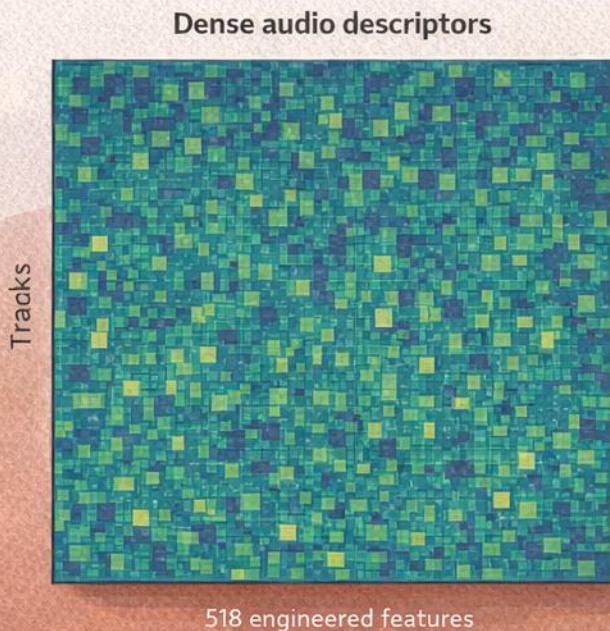


But we standardized the features.



Dense and sparse high-dimensional spaces fail differently

Audio features and lyric vocabularies are both high-dimensional, but not in the same way.



- Dense audio spaces: many coordinates contribute small amounts.
- Sparse lyric spaces: overlap of non-zero terms dominates similarity.
- Sparsity changes the geometry. It does not remove the curse.

Sample complexity explodes

Trying to cover high-dimensional space with fine resolution becomes infeasible quickly.

ϵ	$d = 2$	$d = 5$	$d = 10$
0.5	4	32	1,024
0.2	25	3,125	9.8e+06
0.1	100	100,000	1.0e+10

$$N(\epsilon) \approx \left(\frac{1}{\epsilon}\right)^d$$

$d \uparrow$ or $\epsilon \downarrow \Rightarrow N(\epsilon)$ explodes

A naive grid argument is crude, but it exposes the exponential dependence on dimension immediately.

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Dimensionality reduction is structural, not cosmetic

It is different from merely dropping columns.

Feature selection

- keep tempo
 - keep energy
 - drop lyric_bigram_1421
 - drop embed_998
- choose original coordinates

Dimensionality reduction

- z1 = mix of rhythm + energy
 - z2 = mix of timbre + lyrics
 - new axes preserve structure
 - visualization becomes usable
- construct new coordinates

High-dimensional **disappointment**

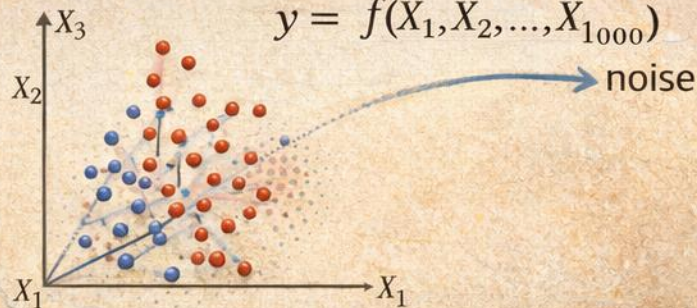
- Precisely. Welcome to high-dimensional disappointment. | So more features can mean less useful signal?

Precisely. Welcome to high-dimensional disappointment.

So more features can mean less useful signal?

$$y = f(X_1, X_2, \dots) + \text{noise}$$

$$y = f(X_1, X_2, \dots, X_{1000})$$

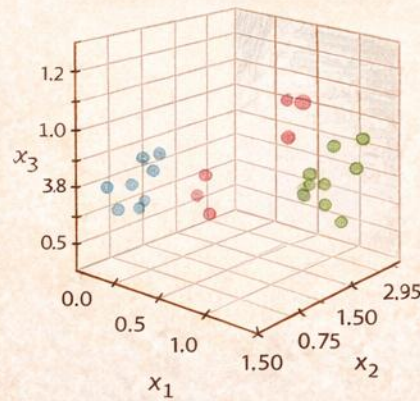


The escape route is a better representation

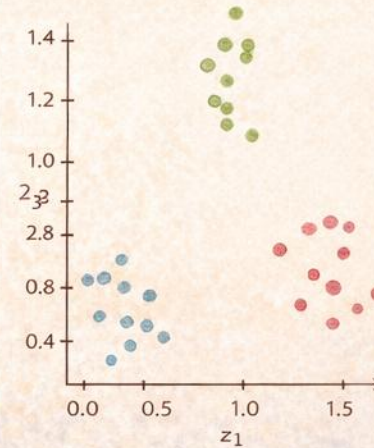
We look for fewer coordinates that preserve the strongest structure.



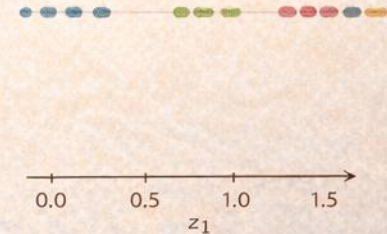
Original space



Reduced to 2D



Even fewer coordinates



High-dimensional points

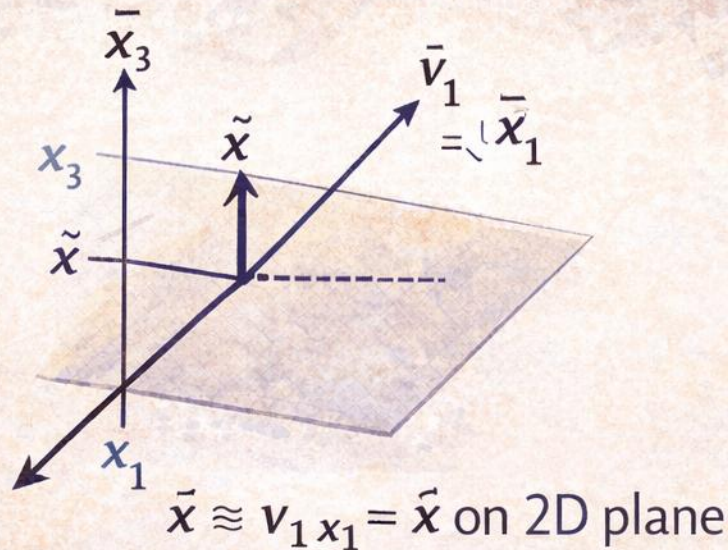
→ lower-dimensional representation

→ more interpretable structure

How do we escape ?

We reduce dimension,
carefully, and with
linear algebra.

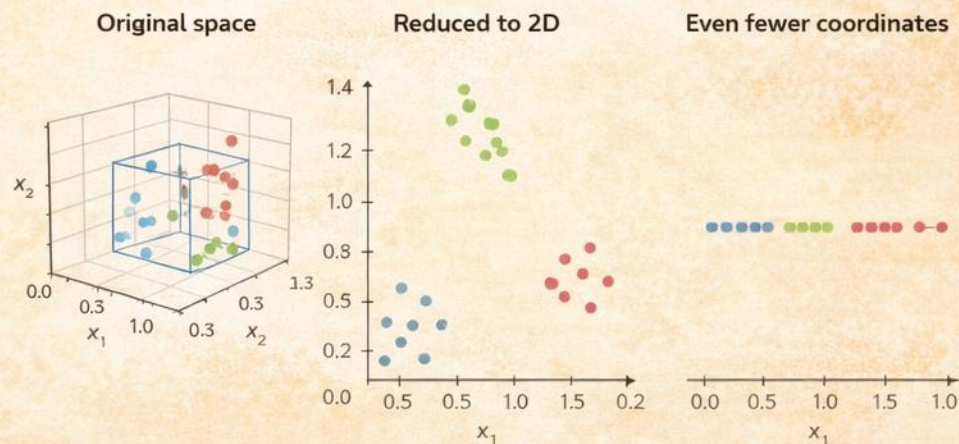
So how do we
escape?



Week 3 takeaways

The theory sets up the PCA lecture that follows next week.

- More dimensions can increase noise, sparsity, and instability.
- The inscribed ball shrinks relative to the cube as d grows.
- Distances grow but become less contrastive.
- Recommendation and clustering need structure beyond raw ambient coordinates.
- Dimensionality reduction creates a workable representation.



Next week: PCA and the geometry of variance.

CODE

SLEEP

REPEAT